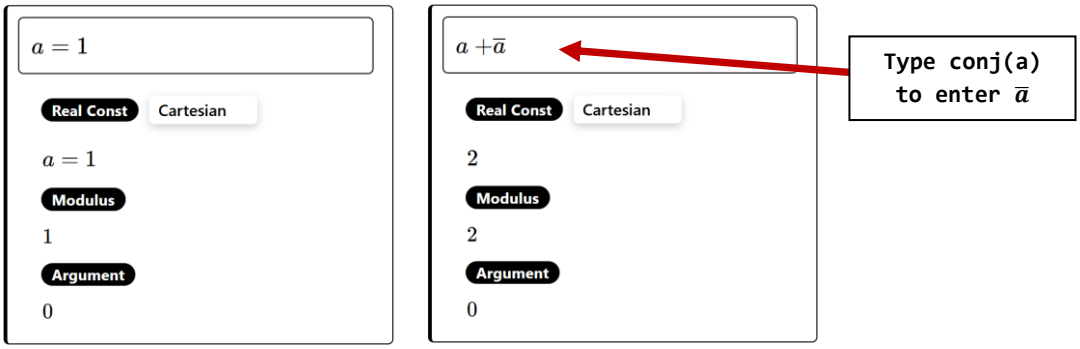
Conjugates through Geometry

- Open **Komplexiti** (https://www.korovatron.co.uk/komplexiti/)
- Add the **constant** $a = 1$ to the panel.
- Add the **expression** $a + \overline{a}$ to a second card.



For each of the following expressions use the mouse to **drag** the point **a** around the Argand diagram and observe what happens.

$a + \overline{a}$	a-$\overline{a}$
Describe how $a + \overline{a}$ behaves as you drag the point a around the Argand diagram.	**Describe how a-$\overline{a}$ behaves as you drag the point a around the Argand diagram.**
Can you prove what you have discovered?	**Can you prove what you have discovered?**

Conjugates through Geometry



$a\bar{a}$	$\frac{a}{\bar{a}}$
Describe how $a\bar{a}$ behaves as you drag the point a around the Argand diagram.	Describe how $\frac{a}{\bar{a}}$ behaves as you drag the point a around the Argand diagram.
Can you prove what you have discovered?	Can you prove what you have discovered?